

# LAB EXPERIMENT RECORD

## Set 10: Survey Responses Analysis

|                           |                                           |
|---------------------------|-------------------------------------------|
| Department                | CSE-DS                                    |
| Course Code / Course Name | DSA0601 / Data Handling and Visualization |
| Experiment                | Data Visualization using R                |
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| Register Number           | 192473018                                 |
| Date                      | 06 October 2026                           |
| Day                       | Tuesday                                   |

### Dataset: Survey Responses

| Survey ID | Question 1 | Question 2 | Question 3 |
|-----------|------------|------------|------------|
| 1         | A          | B          | C          |
| 2         | B          | A          | D          |
| 3         | C          | A          | B          |

### AIM

To analyze survey responses using grouped bar chart, stacked bar chart, and a table, and to develop a Tableau dashboard for interactive exploration of survey responses.

### PROGRAM 1: GROUPED BAR CHART OF QUESTION 1

#### Aim

To create a grouped bar chart to visualize the distribution of answers for Question 1.

#### R Code

R

```
survey <- data.frame(
```

```

Survey_ID = c(1, 2, 3),

Question_1 = c("A", "B", "C"),

Question_2 = c("B", "A", "A"),

Question_3 = c("C", "D", "B")

)

q1 <- table(survey$Question_1)

barplot(q1,

      main = "Distribution of Answers for Question 1",

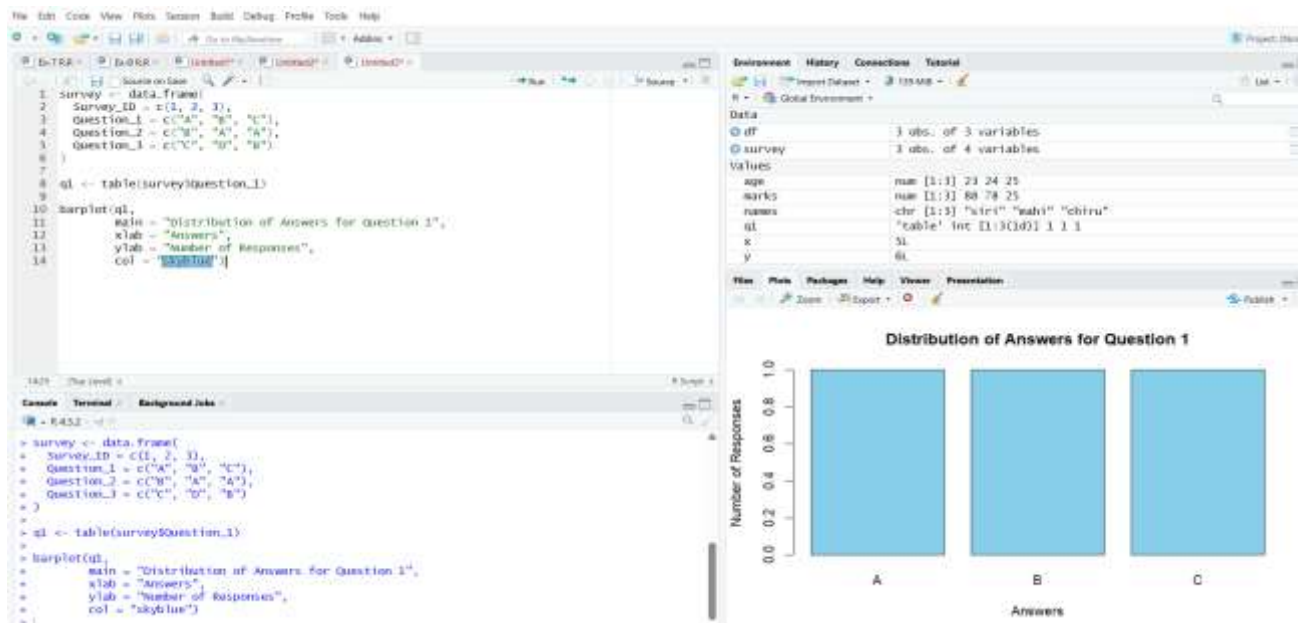
      xlab = "Answers",

      ylab = "Number of Responses",

      col = "skyblue")

```

## Output



Grouped bar chart showing the distribution of answers A, B, and C for Question 1.

## Observation

The chart shows that Question 1 has three different answers. Answer A, B, and C each occur once.

## PROGRAM 2: STACKED BAR CHART OF ALL QUESTIONS

### Aim

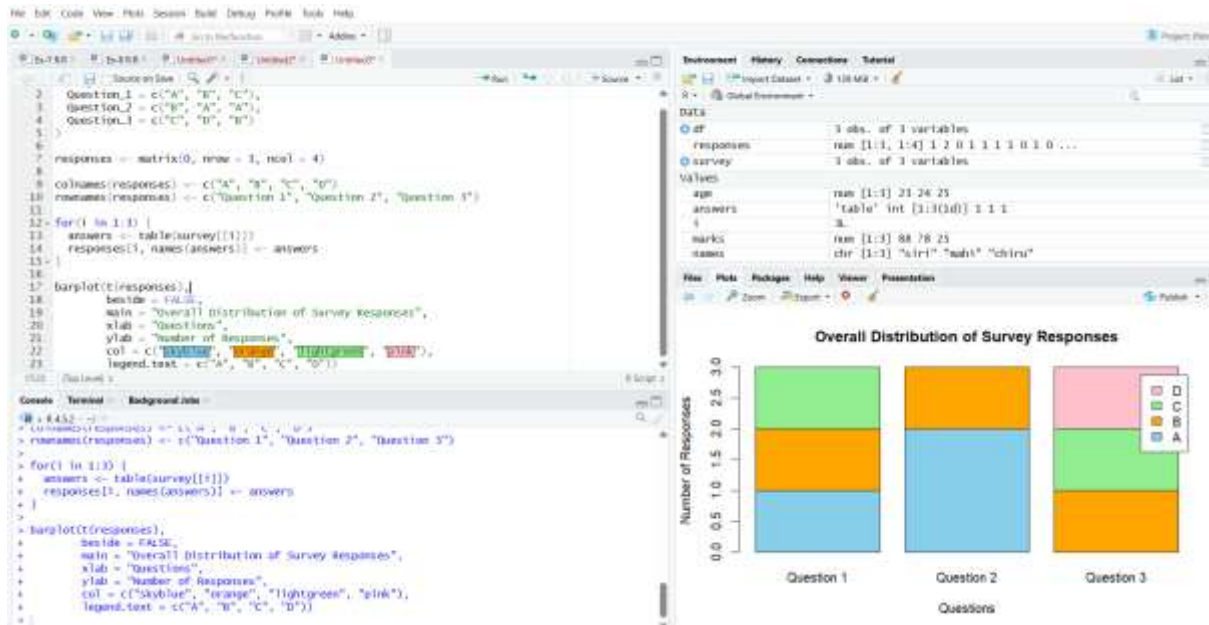
To generate a stacked bar chart representing the overall distribution of responses for all three questions.

### **R Code**

R

```
survey <- data.frame(
  Question_1 = c("A", "B", "C"),
  Question_2 = c("B", "A", "A"),
  Question_3 = c("C", "D", "B")
)
responses <- rbind(
  table(survey$Question_1),
  table(survey$Question_2),
  table(survey$Question_3)
)
responses <- responses[, c("A", "B", "C", "D")]
responses[is.na(responses)] <- 0
barplot(t(responses),
  beside = FALSE,
  main = "Overall Distribution of Survey Responses",
  xlab = "Questions",
  ylab = "Number of Responses",
  col = c("skyblue", "orange", "lightgreen", "pink"),
  legend.text = c("A", "B", "C", "D"))
```

### **Output**



Stacked bar chart showing the overall distribution of answers A, B, C, and D for all three questions.

## Observation

The stacked bar chart compares the responses for all three questions. It shows the number of occurrences of each answer option for every question.

## PROGRAM 3: TABLE OF SURVEY RESPONSES

### Aim

To build a table showing the survey response data for each survey.

### R Code

R

```

survey <- data.frame(

  Survey_ID = c(1, 2, 3),

  Question_1 = c("A", "B", "C"),

  Question_2 = c("B", "A", "A"),

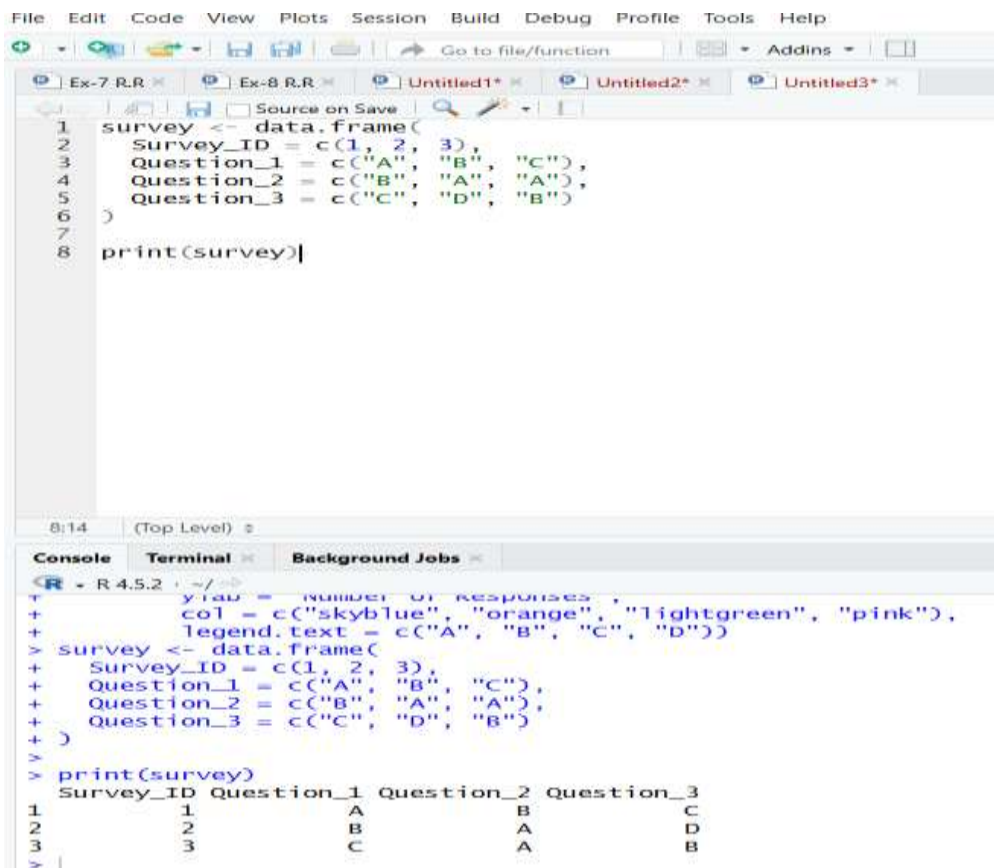
  Question_3 = c("C", "D", "B")

)

print(survey)

```

## Output



The screenshot shows the R Studio environment. The source editor contains the following R code:

```
1 survey <- data.frame(  
2   Survey_ID = c(1, 2, 3),  
3   Question_1 = c("A", "B", "C"),  
4   Question_2 = c("B", "A", "A"),  
5   Question_3 = c("C", "D", "B")  
6 )  
7  
8 print(survey)|
```

The console shows the execution of the code and the resulting data frame:

```
R - R 4.5.2 ~ />  
+ ylab = number of responses ,  
+ col = c("skyblue", "orange", "lightgreen", "pink"),  
+ legend.text = c("A", "B", "C", "D"))  
> survey <- data.frame(  
+   Survey_ID = c(1, 2, 3),  
+   Question_1 = c("A", "B", "C"),  
+   Question_2 = c("B", "A", "A"),  
+   Question_3 = c("C", "D", "B")  
+ )  
> print(survey)  
Survey_ID Question_1 Question_2 Question_3  
1         1         A         B         C  
2         2         B         A         D  
3         3         C         A         B
```

## Observation

The table clearly displays the responses given by each survey participant for all three questions.

## PROGRAM 4: TABLEAU DASHBOARD

### Aim

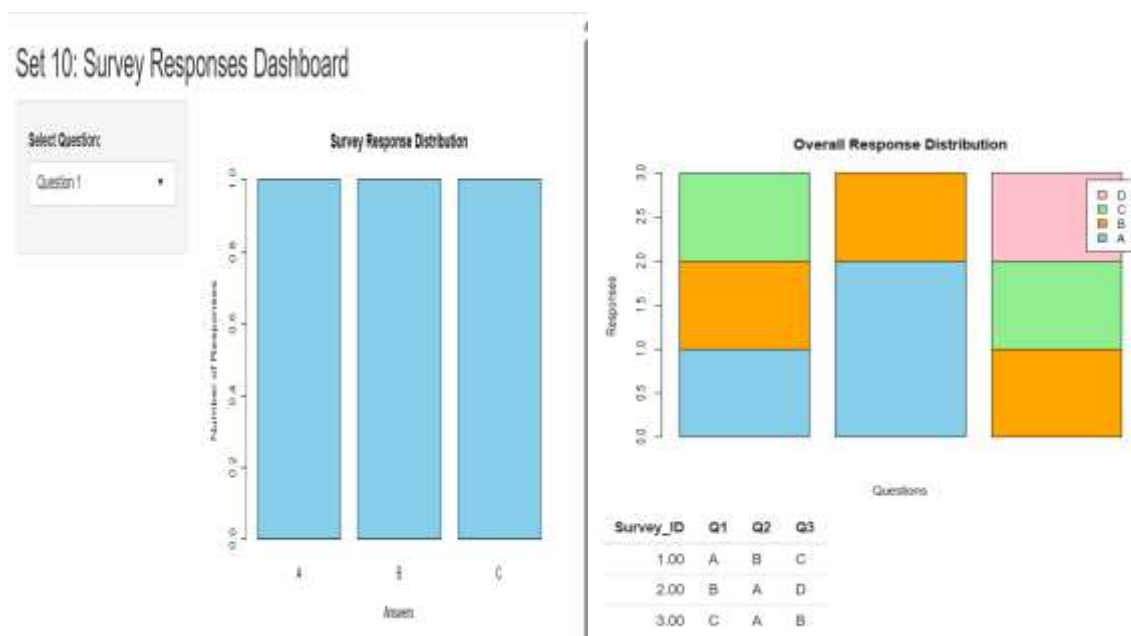
To develop a Tableau dashboard combining the grouped bar chart, stacked bar chart, and survey response table for interactive exploration of survey responses.

### Steps

1. Open Tableau.
2. Select **Microsoft Excel** or **Text File** depending on the saved dataset format.
3. Import the Survey Responses dataset.
4. Create a worksheet for the **Question 1 Bar Chart**.
5. Create another worksheet for the **Stacked Bar Chart**.
6. Create a third worksheet for the **Survey Response Table**.

7. Click **Dashboard** → **New Dashboard**.
8. Drag the Question 1 bar chart into the dashboard.
9. Drag the stacked bar chart into the dashboard.
10. Drag the survey response table into the dashboard.
11. Arrange all three components properly.
12. Add suitable titles and labels.
13. Add filters if required for interactive exploration.
14. Save the Tableau dashboard.

## Output



Interactive Tableau dashboard combining the grouped bar chart, stacked bar chart, and survey response table.

## Observation

The dashboard provides a clear and interactive view of the survey responses. Users can compare Question 1 responses, analyze the overall distribution of answers, and view individual survey responses in the table.